

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary End Semester Examination – Winter 2022-23 Course: B. Tech. Branch: Mechanical Engg. Semester : III Subject Code & Name: Fluid Mechanics (BTMEC303) Max Marks: 60 Date: __/01/2023 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
Q. 1	Describe the following.	(Level/CO)	Marks
A)	Describe any three terms: 1) Mach number, 2) Pascal's Law, 3) Manometer, 4) Steady and unsteady flow, 5) Incompressible fluid flow	Understand	6
B)	i) Calculate the capillary rise in a glass tube of 2.0 mm diameter when immersed vertically in (a) water and (b) Mercury. Take surface tension = 0.08 N/m for water and 0.5 N/m for mercury in contact with air. The specific gravity of mercury is 13.6 and angle of contact is 120°. ii) A hydraulic press has a ram of 25 cm diameter and plunger of 4 cm diameter. It is used to lift a weight of 50 kN. Find the force required at the plunger.	Evaluate	3 × 2
Q.2	Attempt any two questions A/B/C		
A)	A rectangular vertical plane surface is 2 m wide and 2 m deep. Determine the total pressure force and position of centre of pressure on the plane surface when its upper edge is horizontal and coincides with water level. Also determine the force and centre of pressure position when top edge of plane surface is 1.5 m below the water level.	Evaluate	6
B)	Describe the conditions of stability of floating as well as submerged bodies along with neat schematic and tabular presentation.	Understand	6
C)	A rectangular block is 5 m long, 4 m wide and 1.4 m high. The depth of immersion of the block is 0.8 m in the sea water. If the centre of gravity is 0.7 m above the bottom of the block, determine the metacentric height. Assume density of sea water as 1020 kg/cu. m.	Analyze and Evaluate	6
Q. 3	Attempt any two of the following		
A)	Derive the Bernoulli's equation with the help of Euler's equation.	Understand	6
B)	The right limb of a simple U-tube manometer containing mercury is open to the atmosphere while the left limb is connected to a pipe in which a fluid of specific gravity 0.85 is flowing. The centre of the pipe is 0.12 m below the level of mercury in the right limb. Find the pressure of fluid in the pipe if the difference of mercury level in the two limbs is 0.2 m.	Analyze and Evaluate	6
C)	A 0.35 m × 0.18 m Venturimeter is inserted in a vertical pipe of diameter 0.35 m. The throat size is 0.18 m. The water is flowing through the pipe in the upward direction. A differential	Evaluate	6

	mercury manometer connected to the inlet and throat of Venturimeter gives a reading of 0.2 m. Find the flow rate of water (Assume, $C_d = 0.98$).		
Q.4	Attempt any two of the following		
A)	Describe in detail the various types of major and minor losses in pipes.	Understand	6
B)	Explain: i) Dimensional Homogeneity and ii) Raleigh's method	Understand	6
C)	Explain: i) Water hammer effect in pipes, ii) Hydraulic gradient line and Total gradient line	Remember	6
Q. 5	Attempt the following		
A)	Describe Reynolds experiment with the help of neat sketches along with the significance of Reynolds number.	Understand	6
B)	Three pipes of lengths 800m, 600m, and 400 m and of diameters 40 cm, 35cm and 30 cm, respectively are connected in series. These pipes are to be replaced by a single pipe of length 1600 m. Find the diameter of single equivalent pipe.	Analyze and Evaluate	6
	*** End ***		

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